

Linear Temporal Logic on Finite Traces

Game-theoretic Approach to LTLf Planning and Synthesis

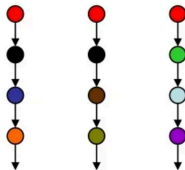
Giuseppe De Giacomo



Temporal logic

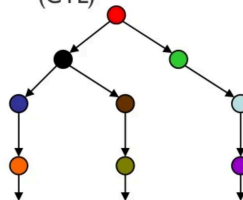
- Linear Time

- Every moment has a unique successor
- Infinite sequences (words)
- Linear Time Temporal Logic (LTL)



- Branching Time

- Every moment has several successors
- Infinite tree
- Computation Tree Logic (CTL)



Artificial Intelligence and in particular the Knowledge Representation and Planning community well aware of temporal logics since a long time:

- Temporally extended goals [BacchusKabanza96]
- Temporal constraints on trajectories [GereviniHslumLongSaettiDimopoulos09 - PDDL3.0 2009]
- Declarative control knowledge on trajectories [BaierMcIlraith06]
- Procedural control knowledge on trajectories [BaierFrizMcIlraith07]
- Temporal specification in planning domains [CalvaneseDeGiacomoVardi02]
- Planning via model checking
 - ▶ Branching time (CTL) [CimattiGiunchigliaGiunchigliaTraverso97]
 - ▶ Linear time (LTL) [DeGiacomoVardi99]

Temporal extended goals and constraints in AI

Foundations borrowed from **temporal logics** studied in CS, in particular:
Linear Temporal Logic (LTL) [Pnueli77].

However:

- Often, LTL is interpreted on finite trajectories/traces.
 - Often, distinction between interpreting LTL on infinite or on finite traces is blurred.
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LTL over finite traces

LTL_f: the language (in symbols)

Same syntax as standard LTL but interpreted over finite traces

$$\varphi ::= A \mid \neg\varphi \mid \varphi_1 \wedge \varphi_2 \mid \varphi_1 \vee \varphi_2 \mid \varphi_1 \supset \varphi_2 \mid \bigcirc\varphi \mid \bullet\varphi \mid \diamond\varphi \mid \square\varphi \mid \varphi_1 \mathcal{U} \varphi_2$$

- A : atomic propositions
- $\neg\varphi, \varphi_1 \wedge \varphi_2, \varphi_1 \vee \varphi_2, \varphi_1 \supset \varphi_2$: boolean connectives
- $\bigcirc\varphi$: “(next step exists and) at next step (of the trace) φ holds”
- $\bullet\varphi$: “if next step exists then at next step φ holds” (weak next) ($\bullet\varphi \equiv \neg\bigcirc\neg\varphi$)
- $\diamond\varphi$: “ φ will eventually hold” ($\diamond\varphi \equiv \text{true} \mathcal{U} \varphi$)
- $\square\varphi$: “from current till last instant φ will always hold” ($\square\varphi \equiv \neg\diamond\neg\varphi$)
- $\varphi_1 \mathcal{U} \varphi_2$: “eventually φ_2 holds, and φ_1 holds until φ_2 does”

LTL_f: the language (in words)

Note: we do not need fancy symbols we can use english words instead:

$$\varphi ::= A \mid \neg\varphi \mid \varphi_1 \wedge \varphi_2 \mid \varphi_1 \vee \varphi_2 \mid \varphi_1 \supset \varphi_2 \mid \text{next } \varphi \mid \text{wnext } \varphi \mid \text{eventually } \varphi \mid \text{always } \varphi \mid \varphi_1 \text{ until } \varphi_2$$

In symbols

$\Diamond A$	"eventually A "	<i>reachability</i>
$\Box A$	"always A "	<i>safety</i>
$\Box(A \supset \Diamond B)$	"always if A then eventually B "	<i>reactiveness</i>
$A \mathcal{U} B$	" A until B "	<i>strong until</i> – stronger than English until
$A \mathcal{U} B \vee \Box A$	" A until B "	<i>weak until</i> – just like English until

In words

<i>eventually A</i>	"eventually A "	<i>reachability</i>
<i>always A</i>	"always A "	<i>safety</i>
<i>always($A \supset$ eventually B)</i>	"always if A then eventually B "	<i>reactiveness</i>
<i>A until B</i>	" A until B "	<i>strong until</i> – stronger than English until
<i>A until $B \vee$ always A</i>	" A until B "	<i>weak until</i> – just like English until

Example

- “All coffee requests from person p will eventually be served”:

$$\Box(\text{request}_p \supset \Diamond \text{coffee}_p)$$

- “Every time the robot opens door d it closes it immediately after”:

$$\Box(\text{openDoor}_d \supset \bigcirc \text{closeDoor}_d)$$

- “Before entering restricted area a the robot must have permission for a ”:

$$\neg \text{inArea}_a \mathcal{U} \text{getPerm}_a \vee \Box \neg \text{inArea}_a$$

Finite Traces

The semantics of LTL_f is given in terms of **finite traces** denoting a finite sequence of consecutive instants of time.

- Finite traces are **finite words** π over the alphabet of $2^{\mathcal{P}}$, i.e., as alphabet we have all the possible propositional interpretations of the propositional symbols in $\mathcal{P}$.
- We denote the **length** of a trace π as $length(\pi)$.
- We denote the **positions**, i.e. instants, on the trace as π, i with $0 \leq i \leq last$, where $last = length(\pi) - 1$ is the last element of the trace.

LTL_f Semantics

Given a finite trace π , we inductively define when an LTL_f formula φ is true at an instant i (for $0 \leq i \leq \text{last}$), in symbols $\pi, i \models \varphi$, as follows:

- $\pi, i \models A$, for $A \in \mathcal{P}$ iff $A \in \pi(i)$.
- $\pi, i \models \neg\varphi$ iff $\pi, i \not\models \varphi$.
- $\pi, i \models \varphi_1 \wedge \varphi_2$ iff $\pi, i \models \varphi_1$ and $\pi, i \models \varphi_2$.
- $\pi, i \models \bigcirc\varphi$ iff $i+1 \leq \text{last}$ and $\pi, i+1 \models \varphi$.
- $\pi, i \models \bullet\varphi$ iff $i+1 \leq \text{last}$ implies $\pi, i+1 \models \varphi$.
- $\pi, i \models \Diamond\varphi$ iff for some j such that $i \leq j \leq \text{last}$, we have $\pi, j \models \varphi$.
- $\pi, i \models \Box\varphi$ iff for all j such that $i \leq j \leq \text{last}$, we have $\pi, j \models \varphi$.
- $\pi, i \models \varphi_1 \mathcal{U} \varphi_2$ iff for some j such that $i \leq j \leq \text{last}$, we have $\pi, j \models \varphi_2$ and for all k , $i \leq k < j$, we have $\pi, k \models \varphi_1$.

Example

Consider the following formula:

$$\Diamond A$$

- On **infinite** traces:
- On **finite** traces:

Example

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- On **infinite** traces:



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Example

Consider the following formula:

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Consider the following formula:

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Example

Consider the following formula:

$$\Diamond \bigcirc A$$

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Example

Consider the following formula:

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Example

Consider the following formula:

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- On **finite** traces:

None!!!

Example

Consider the following formula:

$$\Box \bullet A$$

- On **infinite** traces (in LTL $\bullet A$ must be replaced by $\neg \bigcirc \neg A$ which is equivalent to $\bigcirc A$):
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Consider the following formula:

$$\Box(A \supset \Diamond B)$$

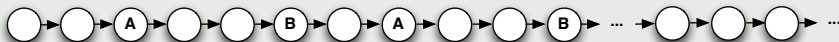
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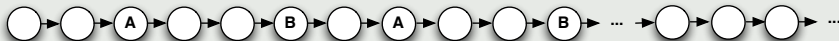
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Example

Consider the following formula:

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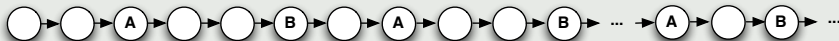
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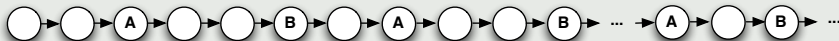
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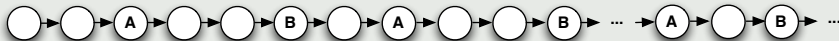
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Consider the following formula:

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- On **finite** traces:

***A* and *B* cannot appear at all!**

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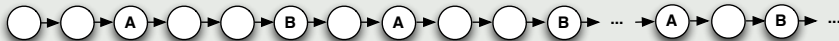
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Example (“Fairness” does not make sense in LTL_f)

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for any point in the trace there is a point later where A holds (“Fairness”).

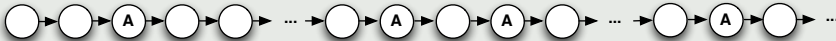
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*In other words, no direct nesting of **eventually** and **always** connectives is meaningful in LTL_f, this contrast what happens in LTL of infinite traces.*

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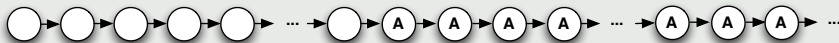
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Example (Capturing STRIPS Planning as LTL_f SAT)

- For each action $A \in Act$ with precondition φ and effects $\bigwedge_{F \in Add(A)} F \wedge \bigwedge_{F \in Del(A)} \neg F$
 - $\square(\bigcirc A \supset \varphi)$: if next action A has occurred (denoted by a proposition A) then now precondition φ must be true;
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- At every step one and only one action is executed: $\square((\bigvee_{A \in Act} A) \wedge (\bigwedge_{A_i, A_j \in Act, A_i \neq A_j} A_i \supset \neg A_j))$.
- Initial situation is described as the conjunction of propositions $Init$ that are true/false at the beginning of the trace: $\bigwedge_{F \in Init} F \wedge \bigwedge_{F \notin Init} \neg F$.
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 - $\square(\bigcirc A \supset \varphi)$: if next action A has occurred (denoted by a proposition A) then now **precondition** φ must be true;
 - $\square(\bigcirc A \supset \bigcirc(\bigwedge_{F \in Add(A)} F \wedge \bigwedge_{F \in Del(A)} \neg F))$: when A occurs, its **effects** are true;
 - $\square(\bigcirc A \supset \bigwedge_{F \notin Add(A) \cup Del(A)} (F \equiv \bigcirc F))$: everything not in add or delete list, remains unchanged.
- At every step one and only **one action** is executed: $\square((\bigvee_{A \in Act} A) \wedge (\bigwedge_{A_i, A_j \in Act, A_i \neq A_j} A_i \supset \neg A_j))$.
- Initial situation** is described as the conjunction of propositions $Init$ that are true/false at the beginning of the trace: $\bigwedge_{F \in Init} F \wedge \bigwedge_{F \notin Init} \neg F$.
- Finally **goal** φ_g eventually holds: $\diamond \varphi_g$.

Thm: A plan exists iff the LTL_f formula is SAT.

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Example (Propositional SitCalc Basic Action Theories in LTL_f)

- Successor state axiom (instantiated for each action A) $F(do(A, s)) \equiv \varphi^+(s) \vee (F(s) \wedge \neg\varphi^-(s))$ can be fully captured:

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- Precondition axioms $Poss(A, s) \equiv \varphi_A(s)$ can **only** be captured in the part saying “if A happens then its precondition must be true”:

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- **No**, $\neg \bigcirc \varphi \equiv \bullet \neg \varphi$ holds in LTL_f !
- Can you denote the last element of the trace in LTL_f ?
- **Yes** $Last \equiv \bullet false$
- Can you talk about the first element in the trace?
- **Yes** e.g. A
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Weak Until

Weak Until, denoted by $\varphi_1 \mathcal{W} \varphi_2$ says that “ φ_1 holds until φ_2 holds, however it is fine for φ_2 not to hold at all, and in that case φ_1 holds forever”. Note this is the typical interpretation of the word “until” in English. Formally it is defined as:

$$\varphi_1 \mathcal{W} \varphi_2 \doteq (\varphi_1 \mathcal{U} \varphi_2) \vee \Box \varphi_1$$

Release

Release denoted by $\varphi_1 \mathcal{R} \varphi_2$ says that “ φ_1 releases φ_2 from holding forever”. Informally, it means that φ_2 must hold until and including the point where φ_1 first becomes true. If φ_1 never becomes true, then φ_2 must remain true until the end of the trace. It can be defined as:

$$\varphi_1 \mathcal{R} \varphi_2 \doteq \varphi_2 \mathcal{W} (\varphi_1 \wedge \varphi_2)$$

The following holds:

- $\varphi_1 \mathcal{R} \varphi_1 \equiv \neg(\neg \varphi_1 \mathcal{U} \neg \varphi_2)$
- it also holds that
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- $\varphi_1 \vee \varphi_2 \equiv \neg(\neg\varphi_1 \wedge \neg\varphi_2)$ and equivalently $\varphi_1 \wedge \varphi_2 \equiv \neg(\neg\varphi_1 \vee \neg\varphi_2)$
- $\bullet\varphi \equiv \neg\bigcirc\neg\varphi$ and equivalently $\bigcirc\varphi \equiv \neg\bullet\neg\varphi$
- $\Box\varphi \equiv \neg\diamond\neg\varphi$ and equivalently $\diamond\varphi \equiv \neg\Box\neg\varphi$
- $\varphi_1\mathcal{R}\varphi_2 \equiv \neg(\neg\varphi_1\mathcal{U}\neg\varphi_2)$ and equivalently $\varphi_1\mathcal{U}\varphi_2 \equiv \neg(\neg\varphi_1\mathcal{R}\neg\varphi_2)$

Negation Normal Form for LTL_f

NNF

Negation Normal Form for LTL_f : for $a \in AP$

$$\varphi ::= \text{true} \mid \text{false} \mid A \mid \neg A \mid \varphi \wedge \varphi \mid \varphi \vee \varphi \mid \bigcirc \varphi \mid \bullet \varphi \mid \blacklozenge \varphi \mid \Box \varphi \mid \varphi \mathcal{U} \varphi \mid \varphi \mathcal{R} \varphi$$

Theorem

Each LTL_f formula φ admits an equivalent in NNF, denoted $nnf(\varphi)$, which is obtained in linear time in the size formula by pushing the negation all in exploiting duals.

Exploit duals through the follow equivalences to push negation all the way in:

- $\neg \neg \varphi \equiv \varphi$
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- $\Diamond\varphi \equiv \varphi \vee \bigcirc(\Diamond\varphi)$ –then choose lfp
- $\Box\varphi \equiv \varphi \wedge \bullet(\Box\varphi)$ –then choose gfp
- $\varphi_1 \mathcal{U} \varphi_2 \equiv \varphi_2 \vee (\varphi_1 \wedge \bigcirc(\varphi_1 \mathcal{U} \varphi_2))$ –then choose lfp
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*(Note: in LTL_f , differently from LTL , $\Box\varphi \equiv \varphi \wedge \bigcirc(\Box\varphi)$ does **not** hold.)*